Bailin Liang

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EDUCATION

Jilin University, Changchun, China **Sept 2024–Jun 2027 (Expected)**
M.S. in Automotive Engineering (**Exam-Free Admission**), College of Automotive Engineering Supervisor: Prof. Junnian Wang
Research Direction: vehicle motion control, corner-module vehicle, and the integration of VLM/VLAs with model-/learning-based control
Awards include **WEICHAI POWER Social Scholarship (Top 1%)**, **Outstanding Student Scholarship** and **Academic Scholarship**.

Jilin University, Changchun, China **Sept 2020–Jun 2024**
B.Eng. in Automotive Engineering, College of Automotive Engineering GPA: 87.89/100 | A-
Awards include **Second Class Scholarship** and **Outstanding Student of the College**.

PUBLICATIONS & PATENTS

Papers

[First Author] Generalized Fault-Tolerant Control for Multi-Wheel Corner-Module Vehicles via Reconfigurable MPC.
IEEE Transactions on Industrial Informatics (JCR **Q1**; IF=9.9), under review.
This work proposes a generalized fault-tolerant control framework for multi-wheel corner-module vehicles, integrating automatic topology identification, reconfigurable MPC, and physical latent-space optimization, with HIL validation of real-time performance and robustness.

[First Author] Nonlinear Shimmy Dynamics Analysis of Highly Integrated In-Wheel Motor Driving Unit with Independent Kingpin Steering for Electric Vehicles.
Vehicle System Dynamics (JCR **Q1**; IF=3.9), under review.
This work proposes a nonlinear shimmy dynamics framework for highly integrated in-wheel motor driving units (HIDUs), combines CX-A and eigenspace-projected shooting methods for bifurcation analysis, develops an FPSA method for efficient parameter evaluation, and reveals parameter-dependent shimmy suppression mechanisms through energy-flow analysis.

[Contributor] A Review of Configuration Classification, Application Control and Future Trends of Highly Integrated Driving Corner Modules for Electric Vehicles.
Automotive Innovation (JCR Q1; IF=7.6), under review.

Patents

As the **first student inventor**, I have authored **9 granted or published invention patents** and participated in **18 granted or published invention patents** in total.

- Kingpin Steering and Omni-Directional Steering Dual-Mode Power-Multiplexing Electric Wheel Corner Module. (U.S. filed; CN120246072A, granted Aug 2025.)
- Distributed Supercapacitor Energy Storage System Based on Electric Wheel Corner Modules. (CN120287823A, granted Oct 2025.)
- Land-Air Power-Multiplexing Flying Car Using Electric Wheel Corner Modules. (CN121492543A, published Feb 2026.)
- A Control Method for a Power-Multiplexed Integrated Drive-Braking Electric Wheel System. (CN120287825A, published Jul 2025.)
- Multi-Link-Braking Integrated Drive-Braking Electric Wheel System. (CN120481601A, published Aug 2025.)
- Power-Multiplexed Integrated Drive-Braking Electric Wheel System. (CN120287824A, published Jul 2025.)
- Highly Integrated Driving Unit with Electronically Adjustable Kingpin Caster Angle. (CN121492543A, published Jul 2025.)
- Electronic Wedge Braking System with a Sprung Actuator. (CN120288017A, published Jul 2025.)
- Multi-Mode Redundant Integrated Drive-Braking Electric Wheel System. (CN120287823A, published Jul 2025.)

RESEARCH & INTERNSHIP

Jilin University – National Key R&D Program of China Changchun, China
Main Contributor, Grant No. 2024YFB2505100 Apr 2025–Present

- Served as a core member of the project **Key Technologies of Highly Integrated Driving Units Based on In-Wheel Motors**, developing high-fidelity MATLAB/Simulink mathematical models and Adams Car multibody dynamics models to support shimmy mechanism analysis and parameter sensitivity studies.
- Designed a coordinated driving-braking-steering shimmy stability control strategy for highly integrated driving units, considering strong nonlinear factors such as suspension-steering coupling, steering clearance, and dry friction.

Dynamics-Aware VLM Planning Framework for Autonomous Vehicles

- Current VLM-based planning methods often overlook variations in vehicle dynamic capabilities across chassis configurations, road conditions, and driving scenarios, which may produce dynamically infeasible or unreliable trajectories. This work proposes a dynamics-aware VLM planning framework that integrates real-time vehicle motion envelopes with closed-loop trajectory feasibility feedback, aiming to bridge high-level semantic planning and dynamically executable vehicle motion while improving reliability and safety.

XPENG Motors, XNGP Team Guangzhou, China
Autonomous Driving R&D Intern Jan 2024–May 2024

- Studied the XNet-XBrain-XPlanner modular end-to-end autonomous driving framework adopted in XPENG's XNGP system.
- Participated in the large-scale data closed loop for urban road tests, takeover events, corner cases, and OTA iterations, focusing on failure-scenario identification, abnormal takeover analysis, and root-cause investigation.

Gspeed Formula Racing Team of JLU Changchun, China
Member of Suspension Group Dec 2020–Dec 2023

- Designed and modeled critical suspension components using CATIA, produced engineering drawings in AutoCAD, and built force-analysis tables for corner-wheel components based on hardpoints and vehicle performance targets.
- Participated in the full cycle of suspension system assembly, maintenance, and tuning.

SKILLS

Application & Coding: MATLAB/Simulink, Python, C/C++, CATIA, AutoCAD. **English:** IELTS 6, CET-6.